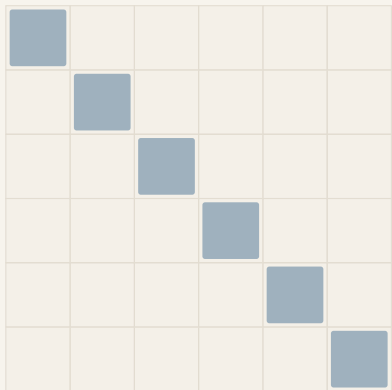


Preconditioner Matrix Structure Comparison

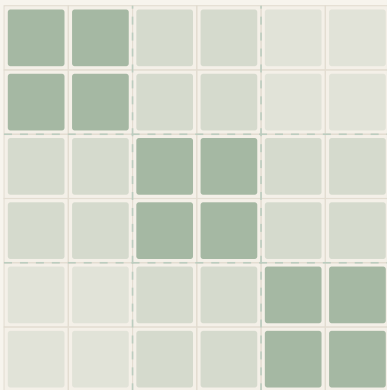
Diagonal AdaGrad



$$\mathbf{P} = \text{diag}(\mathbf{v})$$

$\mathcal{O}(n)$ memory

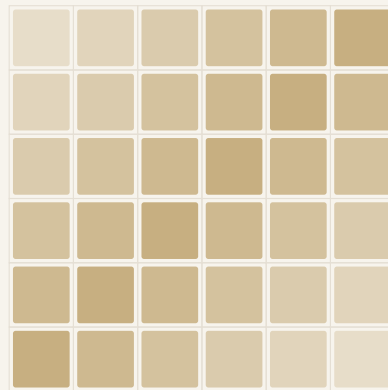
Shampoo ($\mathbf{L} \otimes \mathbf{R}$)



$$\mathbf{P} = \mathbf{L} \otimes \mathbf{R}$$

$\mathcal{O}(m^2 + n^2)$ memory

Full-Matrix AdaGrad



$$\mathbf{P} = (\sum_t \mathbf{g}_t \mathbf{g}_t^\top)^{1/2}$$

$\mathcal{O}(n^2)$ memory